

PYTHON DATA SCIENCE

Essential Commands & Concepts Reference

Version 1.0
2024 Cheat Sheet

N NUMPY ARRAYS

```
import numpy as np

# Create arrays
np.array([1, 2, 3])      # 1D array
np.zeros((3, 3))         # 3x3 zeros
np.ones((2, 4))          # 2x4 ones
np.arange(0, 10, 2)       # [0, 2, 4, 6, 8]
np.linspace(0, 1, 5)      # 5 evenly spaced
```

- ▶ **.shape** → Dimensions (rows, cols)
- ▶ **.reshape()** → Change dimensions
- ▶ **.T** → Transpose array

M MATPLOTLIB / VISUALIZATION

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8, 6))
plt.plot(x, y, label='Line')
plt.scatter(x, y, color='red')
plt.bar(categories, values)
plt.xlabel('X Label')
plt.title('Title')
plt.legend()
plt.show()
```

- ▶ **sns.heatmap()** → Correlation matrix
- ▶ **sns.pairplot()** → Pairwise relationships
- ▶ **sns.distplot()** → Distribution plot

P PANDAS DATAFRAMES

```
import pandas as pd
# Create DataFrame
df = pd.DataFrame({
    'A': [1, 2, 3],
    'B': ['x', 'y', 'z']
})
# Read/Write
pd.read_csv('file.csv') # Read CSV
df.to_csv('out.csv') # Write CSV
```

Method	Description
df.head(n)	First n rows
df.describe()	Statistics summary
df.groupby()	Group by column
df.merge()	Join DataFrames

C DATA CLEANING

```
df.isnull().sum() # Count nulls
df.dropna() # Remove nulls
df.fillna(0) # Fill nulls
df.duplicated() # Find duplicates
df.drop('col', axis=1) # Drop column
df.rename(columns={}) # Rename
df.astype({'col': 'int'}) # Convert type
df['col'].str.lower() # String ops
```

S STATISTICS

Mean: $\bar{x} = (\sum x_i) / n$ | Std: $\sigma = \sqrt{(\sum (x_i - \bar{x})^2) / n}$

```
df['col'].mean() # Mean
df['col'].std() # Standard deviation
df['col'].corr(other) # Correlation
df.quantile([0.25, 0.75]) # Quartiles
```

ML SCIKIT-LEARN WORKFLOW

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

# 1. Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# 2. Scale features
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# 3. Train model
model = RandomForestClassifier(n_estimators=100)
model.fit(X_train, y_train)

# 4. Predict & Evaluate
predictions = model.predict(X_test)
accuracy = accuracy_score(y_test, predictions)
```

A ALGORITHMS QUICK REF

Algorithm	Use Case	Class
LinearRegression	Continuous prediction	sklearn.linear_model
LogisticRegression	Binary classification	sklearn.linear_model
RandomForest	Ensemble, non-linear	sklearn.ensemble
SVM	High dimensional	sklearn.svm
KMeans	Clustering	sklearn.cluster

PRO TIPS

- ▶ Use `df.iloc[]` for position-based indexing, `df.loc[]` for label-based
- ▶ Vectorization with NumPy is 10-100x faster than Python loops
- ▶ `.copy()` to avoid SettingWithCopyWarning in Pandas
- ▶ Set `random_state` for reproducible ML results
- ▶ Use `Pipeline` to prevent data leakage during preprocessing

```
# Jupyter magic commands
%matplotlib inline      # Display plots inline
%timeit function()       # Time execution
%load_ext autoreload     # Auto-reload modules
```

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''' # Save to file with open('/mnt/kimi/output/python_data_science_cheatsheet.html', 'w', encoding='utf-8') as f: f.write(html_content) print("✅ Cheat sheet created successfully!") print("📄 File saved to: /mnt/kimi/output/python_data_science_cheatsheet.html") print("\n 🖨️ Print Instructions:") print(" 1. Open the HTML file in Chrome/Edge") print(" 2. Press Ctrl+P (or Cmd+P on Mac)") print(" 3. Set paper size to A4, landscape orientation") print(" 4. Enable 'Background graphics' in print settings") print(" 5. Set margins to 'None' or minimum (0.4in)") print(" 6. Save as PDF")